

A Multi-Agent LLM Framework for Large-Scale Climate Data Analysis

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Abstract. This study explores the use of Large Language Models (LLMs) to support automated analysis and reporting of large-scale climate datasets, focusing on historical meteorological records from Brazil. The growing availability of climate data presents significant challenges related to data volume, preprocessing, and interpretation, particularly when attempting to extract meaningful long-term patterns from heterogeneous datasets. To address these challenges, this work proposes a pipeline that combines large-scale climate data processing with LLM-based multi-agent analysis. The proposed approach includes the automated collection of historical meteorological data from the Brazilian National Institute of Meteorology (INMET), followed by extensive preprocessing, aggregation, and data reduction techniques designed to manage the scale and noise of the raw observations. The resulting datasets are analyzed using specialized LLM agents that interpret climate trends and identify relevant patterns, such as temperature anomalies and potential heatwave events. Experimental results indicate an overall warming tendency in the aggregated temperature series. The LLM agents contributed by contextualizing candidate anomalies, organizing temporal trends, identifying limitations in the available evidence, and consolidating specialized analyses into structured reports. By integrating large-scale data processing with generative AI-based interpretation, this study contributes to the development of assisted climate analysis workflows, while recognizing that statistical validation and domain-expert assessment remain necessary for confirming climate events and supporting decision-making.

Keywords: Large Language Models · Climate Reporting Automation · Exploratory Trend Analysis · Temperature Trends · Multi-Agent Systems.

1 Introduction

Climate change has long been recognized as a critical global challenge, with rising greenhouse gas concentrations contributing to increasing global temperatures and changes in precipitation patterns and extreme weather events [22, 16, 19]. These transformations affect multiple sectors, including agriculture, energy consumption, and public health. In Brazil, where agribusiness plays a central role in the national economy, variations in temperature and precipitation may

significantly affect agricultural productivity and food systems [3, 9]. Understanding and anticipating these climatic dynamics is therefore essential for supporting sustainable planning and mitigation strategies.

Advances in Artificial Intelligence (AI) and Machine Learning (ML) have opened new possibilities for analyzing large-scale environmental datasets and improving climate forecasting. ML techniques have been successfully applied to tasks such as crop yield prediction, climate modeling, and environmental monitoring, enabling more accurate predictions and more efficient resource management [18]. However, the growing volume and complexity of climate data create new challenges for interpreting results and translating them into actionable insights for decision-makers.

In this context, Large Language Models (LLMs) present an opportunity to support the interpretation and synthesis of complex analytical outputs. By combining predictive data analysis with automated report generation, LLM-based systems can help transform raw climate data into structured insights that are more accessible to policymakers, researchers, and practitioners.

This study proposes a framework for automated climate data analysis and reporting focused on Brazil. The approach integrates three main stages: (i) large-scale collection and preprocessing of historical meteorological data, (ii) data aggregation and trend analysis, and (iii) the use of specialized LLM-based agents to interpret the results and generate structured climate reports. By combining data processing pipelines with multi-agent LLM analysis, the proposed system aims to facilitate the interpretation of large-scale climate data and support evidence-based decision-making in climate-sensitive sectors.

2 Related Works

AI and ML techniques have increasingly been applied to climate research, enabling improved forecasting and data-driven mitigation and adaptation strategies. Several studies integrate ML with traditional physical climate models to enhance predictive performance. For example, Eyring *et al.* [10] highlight the use of hybrid modeling approaches and ML-based emulators to improve projections of extreme climate events, while O’Gorman *et al.* [17] employ Random Forest models to parameterize moist convection in general circulation models. Similarly, Chantry *et al.* [4] demonstrate how ML techniques can support bias correction, downscaling, and parameterization in climate simulations.

Recent research also explores hybrid and ensemble learning approaches for environmental prediction tasks. Ardabili *et al.* [2] provide a comprehensive review of ML and Deep Learning (DL) applications in hydrological and climate-related modeling, emphasizing the growing adoption of ensemble methods to improve forecasting robustness. Li *et al.* [14] propose a hybrid classification–regression framework to enhance predictions of variables such as sea surface temperature.

Beyond climate modeling itself, AI techniques have been applied to broader climate-related challenges, including policy analysis, environmental monitoring, and agricultural forecasting. Duan *et al.* [7] incorporate climatic and economic

variables into Integrated Assessment Models to support climate mitigation strategies, while Rolnick *et al.* [20] discuss a wide range of ML applications for climate change mitigation. In the agricultural domain, Crane-Droesch *et al.* [5] combine statistical and neural network models to predict crop yields under changing climate conditions, and Davenport *et al.* [6] employ Convolutional Neural Networks to detect atmospheric circulation patterns associated with extreme precipitation events.

Complementary technologies have also emerged to improve access to and interpretation of climate data. Zepner *et al.* [24] introduce an interactive platform for visualizing historical climate datasets, while AlQallaf *et al.* [1] explore the use of extended reality (XR) technologies to enhance public awareness of climate change impacts.

Despite these advances, most existing studies focus either on improving predictive models or on specific climate-related applications. Less attention has been given to automating the interpretation of large-scale climate datasets and generating structured analytical reports from complex climate data. In this context, the present work proposes an LLM-based multi-agent approach to automate the analysis and reporting of climate data, with a focus on the Brazilian context.

3 Methodology

This study proposes an exploratory computational pipeline for the automated processing and interpretation of large-scale meteorological data. The approach combines deterministic data-processing procedures with LLM-based agents responsible for interpreting reduced representations of the historical dataset and generating structured analytical reports. As illustrated in Fig. 1, the methodology is organized into three main phases: (i) large-scale collection and preprocessing of meteorological records; (ii) temporal aggregation and data reduction to produce representations compatible with computational and context-window constraints; and (iii) analysis of the resulting summaries by specialized LLM-based agents. The numerical preprocessing and aggregation operations were performed before the agent-based stage. Therefore, the agents did not directly process the complete raw hourly dataset, but rather analyzed previously prepared summaries from distinct analytical perspectives. Each phase is described in detail in the following subsections.

3.1 Data Collection

Meteorological data were collected from the Historical Meteorological Data repository of the Instituto Nacional de Meteorologia (INMET) [12]. The repository provides yearly compressed files, each containing hourly meteorological records for multiple Brazilian cities. To enable large-scale acquisition, a web-scraping script was developed to automatically access the INMET portal, identify available files, and download them to a local directory.

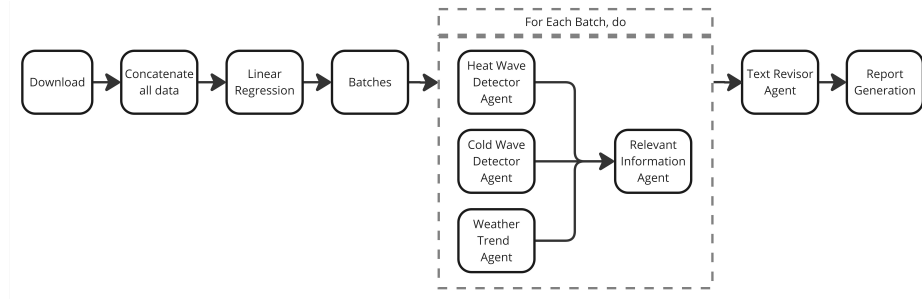


Fig. 1. The Solution design.

Each downloaded `.zip` file contains multiple `.csv` files, each corresponding to a city and a year of observations. The collected data include hourly measurements of temperature, precipitation, atmospheric pressure, humidity, wind, and radiation, together with metadata such as city, state, latitude, longitude, and station altitude. Table 1 summarizes the main meteorological variables considered in this study.

In addition to meteorological variables, the dataset also includes metadata such as city, state, latitude, longitude, and station altitude, which were preserved to support geographically contextualized analyses.

3.2 Data Preparation and Reduction

After collection, the meteorological records were preprocessed to improve data consistency and reduce the computational requirements of the subsequent analyses. The preprocessing pipeline included handling invalid values, missing observations, potential outliers, numerical precision reduction, and temporal aggregation.

Values such as `-9999.0`, `-999.0`, `-99.0`, and `9999.0`, which represent invalid or unavailable measurements in the original files, were converted into missing values. This transformation prevented these codes from being interpreted as valid meteorological observations during aggregation.

Missing values were handled using two complementary strategies implemented with the Pandas library. Short gaps of up to six consecutive hours were filled using forward fill, whereas longer numerical gaps, limited to a maximum interval of 24 hours, were treated through linear interpolation. For categorical attributes, mode imputation was employed. Although these procedures were intended to preserve temporal continuity, they introduce assumptions that may influence the resulting series. Forward fill assumes short-term persistence and may artificially extend previously observed conditions, whereas linear interpolation assumes gradual variation and may smooth abrupt temperature changes. Mode imputation may also overrepresent the most frequent category and reduce local variability. Consequently, these procedures may attenuate short-lived extremes and reduce the variance observed in the processed data.

Table 1. Dictionary of Meteorological Parameters

Column Name	Description	Unit
Date	Represents the date of the measurement in the format year-month-day.	YYYY-MM-DD
Time	Indicates the time of the measurement in UTC.	UTC
Total Precipitation, Hourly	Total accumulated precipitation in millimeters during the specified hour.	millimeters
Atmospheric Pressure at Station Level, Hourly	Atmospheric pressure measured at station level, in millibars.	millibar
Max Atmospheric Pressure Previous Hour	Maximum atmospheric pressure recorded in the previous hour.	millibar
Min Atmospheric Pressure Previous Hour	Minimum atmospheric pressure recorded in the previous hour.	millibar
Global Radiation	Global radiation accumulated during the hour, measured in kilojoules per square meter.	KJ/m ²
Air Temperature - Dry-bulb, Hourly	Air temperature measured by the dry bulb thermometer, in degrees Celsius.	°C
Dew Point Temperature	Dew point temperature during the specified hour, in degrees Celsius.	°C
Max Air Temperature Previous Hour	Maximum air temperature recorded in the previous hour.	°C
Min Air Temperature Previous Hour	Minimum air temperature recorded in the previous hour.	°C
Max Dew Point Temperature Previous Hour	Maximum dew point temperature recorded in the previous hour.	°C
Min Dew Point Temperature Previous Hour	Minimum dew point temperature recorded in the previous hour.	°C
Max Relative Humidity Previous Hour	Maximum relative humidity recorded in the previous hour.	%
Min Relative Humidity Previous Hour	Minimum relative humidity recorded in the previous hour.	%
Relative Humidity, Hourly	Relative humidity measured during the specified hour.	%
Wind Direction Hourly	Predominant wind direction during the specified hour, in degrees.	°
Max Wind Gust	Maximum wind gust speed recorded during the specified hour, in meters per second.	m/s
Wind Speed Hourly	Average wind speed during the specified hour, in meters per second.	m/s

Potential outliers were treated using the Interquartile Range method. Observations below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were classified as anomalous [13, 21]. Although this procedure reduces the influence of possible measurement errors, meteorological datasets may contain physically plausible extreme observations that fall outside these statistical thresholds. The method may therefore remove genuine climatic extremes, particularly in skewed or seasonally heterogeneous distributions, reducing the observed variance and affecting the estimated frequency and intensity of candidate temperature anomalies. Because alternative imputation and outlier-handling strategies were not systematically compared, the robustness of the identified patterns to these preprocessing decisions remains uncertain.

To improve memory efficiency, numerical columns originally represented as 64-bit values were, when applicable, converted to 32-bit values. The records were also processed in batches, preventing the complete historical dataset from being loaded into memory simultaneously.

The hourly observations were initially grouped into daily summaries. Mean values were used for numerical variables, whereas the mode was used for categorical attributes. The aggregation was performed according to the available temporal and station-level identifiers, including date, city, state, and meteorological station metadata.

The daily records were subsequently organized into broader temporal representations, including five-year, ten-year, and complete historical datasets. For the agent-based stage, the temperature data were further summarized into monthly averages. This reduction substantially decreased the number of records provided to the language models while preserving the main temporal patterns required for exploratory interpretation.

The resulting reduced datasets were therefore not intended to reproduce every local or hourly meteorological variation. Instead, they provided computationally manageable summaries of historical records to examine broad temperature behavior and generate structured analytical reports.

3.3 Trend Preparation for Exploratory Analysis

The initial research design considered the implementation of predictive time-series models, including SARIMA, LSTM, and Temporal Fusion Transformer (TFT) [8, 23, 15]. These models were intended to represent seasonal and nonlinear dependencies in the meteorological records. However, due to computational and technical constraints, they were not fully implemented or experimentally evaluated.

Consequently, the final analysis focused on temporal aggregation and exploratory trend representation rather than on developing a validated forecasting model. The historical temperature records were summarized into monthly and annual averages, allowing the visualization of broad variations across the analyzed period.

A moving-average representation was used to reduce short-term fluctuations and highlight the general temporal behavior of the aggregated temperature se-

ries. Values for the period from 2025 to 2050 were included as trend-based extensions of the historical series. These future values should be interpreted as exploratory projections derived from the observed aggregated trend and not as validated climate forecasts.

This distinction is important because the proposed framework does not incorporate physical climate models, probabilistic forecasting, confidence intervals, or a formal evaluation of predictive accuracy. The projected values were used primarily to examine how the LLM agents interpreted a simplified continuation of the historical temperature pattern.

The summarized historical records and trend-based values were subsequently organized into tabular representations. These representations served as inputs to the specialized agents described in the following subsection.

3.4 LLM-Based Multi-Agent Analysis

The final stage of the proposed framework employed multiple LLM-based agents to interpret the reduced climate data and generate structured analytical outputs. The overall agent workflow is illustrated in Fig. 2. Because the complete hourly dataset exceeded the context-window limits of the selected language models, the agents received aggregated tabular representations rather than the original meteorological records.

The input provided to the agents contained summarized temperature values covering the historical period and the exploratory trend-based extension. The records were divided into batches of approximately 200 rows to maintain computational feasibility and avoid exceeding the input capacity of the models. Each batch was processed independently before the generated observations were consolidated.

The framework included three specialized analytical roles:

- a high-temperature analysis agent, responsible for examining unusually high values and identifying periods that could represent candidate temperature anomalies;
- a low-temperature analysis agent, responsible for examining unusually low temperature patterns and indicating when the available information was insufficient for a more specific meteorological interpretation;
- a climate-trend analysis agent, responsible for describing broad temporal patterns and changes in the aggregated temperature series.

The term “candidate anomaly” is used because the agents interpreted patterns in aggregated temperature data without applying a standardized climatological definition of extreme events. In particular, the identification of a potential heat-wave event would require additional information, including local climatological baselines, daily duration, percentile-based thresholds, and geographic context. Therefore, the outputs produced by the agents were treated as exploratory interpretations rather than definitive classifications of meteorological events.

After the specialized agents completed their analyses, their outputs were passed to an intermediate reviewer agent. The reviewer agent was responsible

for consolidating the observations, removing repeated statements, and organizing the most relevant findings into a structured analytical summary. This stage was intended to improve the coherence of the final output rather than to provide an independent statistical validation of the findings.

Three instruction-tuned language models were used in the framework: gemma2-9b-it, llama-3.3-70b-versatile, and mixtral-8x7b-32768, which provide context windows of 8,192, 128,000, and 32,768 tokens, respectively. The study evaluated the multi-agent workflow as an integrated system and did not conduct a controlled comparison of alternative model-task assignments or isolate the contribution of each model. Therefore, the findings should be attributed to the overall multi-agent configuration rather than to the individual capabilities of a specific language model. Future work should systematically compare different model assignments using identical inputs, prompts, and evaluation criteria.

The agent-based reasoning process can therefore be summarized as follows:

1. the preprocessed climate dataset was converted into reduced temporal summaries;
2. the summarized records were divided into batches of approximately 200 rows;
3. each batch was provided to agents with distinct analytical responsibilities;
4. each agent generated an interpretation according to its assigned perspective;
5. the reviewer agent consolidated the specialized outputs;
6. the consolidated observations were organized into the final analytical report.

This workflow separates data processing from language-based interpretation. The statistical preparation of the dataset occurred before the LLM stage, while the agents were used primarily to identify, describe, and communicate patterns present in the summarized data.



Fig. 2. Overview of the multi-agent workflow for climate data interpretation and report generation.

4 Results

This section presents the results obtained from the proposed pipeline for climate data processing and LLM-based interpretation. The results include the outputs of the data extraction and preprocessing stages, the preparation of aggregated datasets for analysis, and the insights generated by the LLM agents when interpreting the processed climate data.

4.1 Data Extraction

The data extraction process was completed successfully; however, it required substantial processing time due to the large volume of meteorological records collected. The dataset combines 25 years of hourly meteorological observations, covering the period from 2000 to 2024, from multiple regions across Brazil. Due to its scale, the extraction and consolidation steps had to be performed in batches, enabling the dataset to be processed without exceeding memory limits.

During this process, several optimizations were applied to improve computational efficiency. Numerical precision was reduced from `float64` and `int64` to `float32` and `int32`, and the original hourly observations were aggregated into daily records. Outliers were also removed during preprocessing, following the IQR-based filtering described in the methodology.

4.2 Data Treatment and Modeling Challenges

Despite the successful extraction of the meteorological records, several challenges emerged during the data treatment phase. The most significant issue was the extremely large volume of data generated by concatenating historical observations across multiple years and stations. In addition, the dataset contained numerous invalid values, missing observations, and extreme outliers that required careful preprocessing before further analysis could be performed.

The dataset size also imposed computational constraints on traditional machine learning workflows, as the full dataset could not be loaded into memory at once. To address these limitations, a combination of batch processing, distributed computation with Dask [11], and reduced numerical precision was employed. As a result, a large portion of the effort in this study was devoted to preprocessing and preparing the data for downstream analysis by the LLM agents.

4.3 Data Output Files

After preprocessing and aggregation, several structured datasets were generated to support further analysis. These datasets were organized into different temporal segments to facilitate both long-term and short-term examination of the climate records. The following datasets were produced:

- **Five-Year Data Files:** datasets segmented into five-year intervals (2000–2004, 2005–2009, 2010–2014, 2015–2019, and 2020–2024).

- **Ten-Year Data Files:** datasets grouped into broader ten-year periods (2000–2009, 2010–2019, and 2020–2024).
- **Comprehensive Data File:** a single consolidated dataset containing all processed records from 2000 to 2024.
- **Sample Data File:** a reduced dataset containing 100 randomly selected observations for each city, state, and date, used primarily for exploratory analyses and testing.

These files enabled flexible exploration of the processed climate data while maintaining manageable dataset sizes for subsequent analysis.

4.4 Adaptation Due to Technical Constraints

The initial research design proposed using the entire historical dataset to train predictive models that forecast future climate variables. However, technical limitations related to dataset size and computational resources required an adaptation of the experimental strategy.

Instead of training predictive models on the full dataset, an aggregated representation of the historical temperature records was used as input for the LLM agents. This adjustment allowed the system to focus on interpreting summarized climate patterns rather than performing full-scale predictive modeling. The resulting dataset, therefore, served as the primary analytical input for the agent-based interpretation stage described in the following subsection.

4.5 Interpreting Results with Agents

The processed climate data were analyzed using the LLM-based multi-agent system described in the methodology. Fig. 3 presents the historical temperature series used as input for the agents.

As shown in Fig. 3, the historical temperature data exhibit a clear upward trend over the analyzed period, although short-term variability introduces significant fluctuations. To better highlight long-term patterns, a moving average representation was used for visualization purposes. Fig. 4 illustrates the resulting smoothed temperature series.

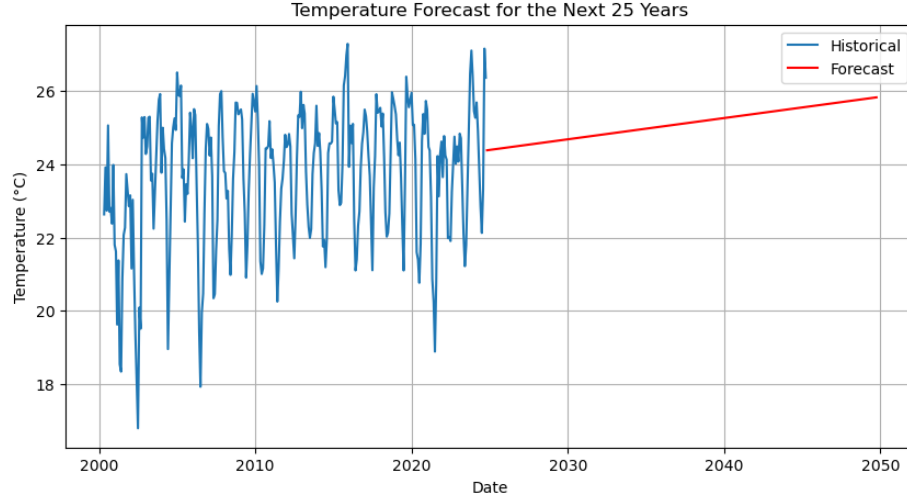


Fig. 3. Temperature forecast from the processed climate dataset.

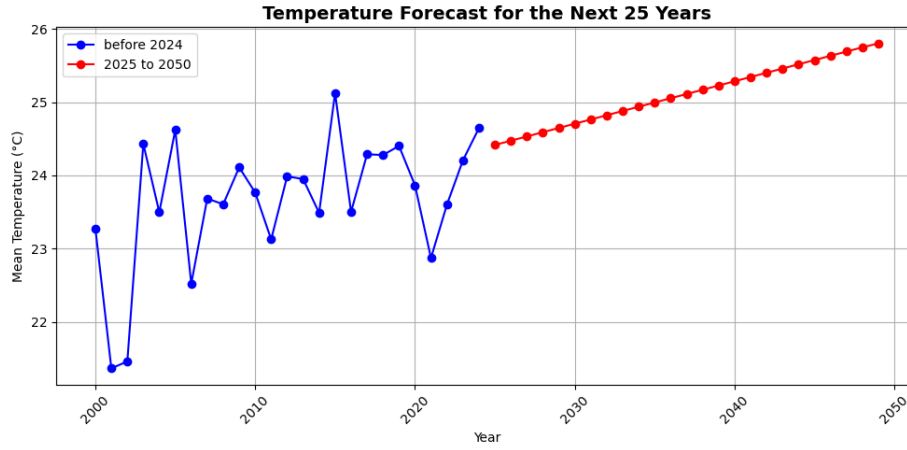


Fig. 4. Moving average representation of historical temperature values.

The distribution of the aggregated temperature values is shown in Fig. 5. The histogram indicates that most observations are concentrated within the higher portion of the analyzed temperature range. Because the histogram does not preserve the temporal ordering of the observations, it is used only to describe the overall distribution and cannot independently demonstrate an increase in higher temperatures over time.

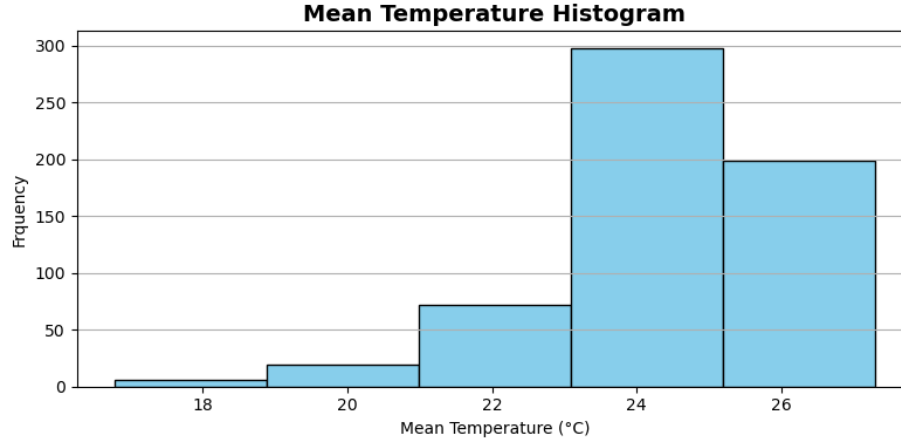


Fig. 5. Histogram of aggregated temperature observations.

Based on these inputs, three representative instances illustrate the specific contribution of the LLM agents to the analytical process. First, the heat-wave detection agent highlighted unusually high September temperature values for 2019, 2023, and 2024, as well as a projected anomaly for 2029. The identified value of approximately 24.543°C was about 2.543°C above the historical September baseline used in the analysis. The agent’s contribution was not to calculate this deviation, which was already present in the aggregated input, but to contextualize it as a candidate temperature anomaly requiring further climatological assessment. These instances cannot be considered confirmed heat waves because the analysis did not apply station-level climatological thresholds or duration-based heat-wave criteria.

Second, the cold-wave detection agent was unable to produce a conclusive classification based solely on the available temperature series. Instead, it identified that atmospheric pressure, wind direction, wind speed, and humidity would be required for a more reliable assessment. In this instance, the agent contributed by identifying the insufficiency of the available evidence and specifying the additional variables needed, rather than producing an unsupported interpretation.

Third, the climate-trend agent segmented the temporal series into six analytical windows and organized the aggregated data into comparisons of temperature variations, coldest years, warmest years, and overall tendencies. The reviewer agent then consolidated the outputs generated by the specialized agents, removed redundant observations, and organized the principal findings into the structured summary presented in Table 2.

These instances indicate that the principal contribution of the LLM agents lies in contextualizing numerical results, assessing whether the available evidence is sufficient for interpretation, and converting multiple analytical outputs into a structured report. The agents did not independently calculate the underlying

trends or validate the identified climate events, which remained dependent on the preprocessing, aggregation, and trend-estimation procedures applied before the agent-based analysis.

Table 2. Historical Temperature Trends and Exploratory Extensions (2000–2050)

Period	Important Facts	Coldest Year	Warmest Year
2000 - 2008	There was a pronounced upward trend in temperatures during this period. Both the maximum and minimum values reached their highest recorded levels, resulting in a temperature variation of up to 3.2°C.	2001 (21.05°C)	2007 (24.75°C)
2008 - 2016	A consistent average increase of 0.1°C was detected in the moving average of the years during this period.	2011 (23.05°C)	2015 (25.05°C)
2016 - 2024	A significant thermal variation was observed over the three-year span from 2021 to 2024, with 2022 and 2023 each recording progressively higher average temperatures, culminating in even greater values by 2024.	2021 (22.89°C)	2024 (24.70°C)
2024 - 2033	Upward trend estimated from aggregated series.	2025 (24.50°C)	2033 (24.90°C)
2033 - 2042	Upward trend estimated from aggregated series.	2034 (24.95°C)	2042 (25.35°C)
2042 - 2050	Upward trend estimated from aggregated series.	2042 (25.40°C)	2050 (25.80°C)

Overall, the segmented analysis indicates an upward tendency in the aggregated temperature series and highlights several candidate temperature anomalies identified by the agents.

5 Discussion

The comparative analysis indicates that although both this study and previous work address important aspects of climate forecasting using machine learning methodologies, there are clear differences in scope, analytical focus, and regional emphasis. Many existing studies concentrate on specific climate-related phenomena or geographically limited contexts, applying machine learning models to solve narrowly defined prediction tasks. For example, Davenport *et al.* [6] employ convolutional neural networks (CNNs) to predict extreme precipitation events in the U.S. Midwest, achieving high accuracy in detecting localized atmospheric circulation patterns. Similarly, Crane-Droesch *et al.* [5] propose a semiparametric neural network (SNN) designed to model crop yields under extreme weather conditions, emphasizing agricultural impacts rather than broader climatic dynamics. Chantry *et al.* [4], in turn, focus on applying machine learning techniques

for bias correction and downscaling in meteorological models, addressing technical challenges related to climate simulation across different temporal scales.

From a methodological perspective, the approach proposed in this work shares certain conceptual similarities with studies such as Eyring *et al.* [10], which advocate hybrid strategies that combine statistical and physical modeling techniques to improve predictive reliability. However, the use of large language models for interpretative analysis introduces an additional analytical layer that distinguishes the present approach. By incorporating LLM-based agents that can analyze aggregated climate data, the proposed framework enhances the interpretability and accessibility of large meteorological datasets. Furthermore, the segmentation of data into five-year and ten-year intervals, combined with the use of moving averages to identify long-term patterns, follows analytical practices commonly adopted in climate research while adapting them to a large-scale regional dataset.

Regarding the observed outcomes, previous studies have demonstrated strong performance in addressing specific forecasting challenges. Davenport *et al.* [6], for instance, report high accuracy in the classification of extreme precipitation events, while Crane-Droesch *et al.* [5] show significant reductions in prediction errors within agricultural modeling contexts. Nevertheless, these results are generally constrained to localized phenomena or domain-specific applications. In contrast, the present study focuses on broader climatic patterns and long-term trends within the Brazilian context, aiming to provide a more comprehensive perspective on regional climate dynamics. The identification of persistent warming trends and recurring temperature anomalies, including potential heat wave events, contributes to understanding how climate change may manifest in a region of considerable ecological and socio-economic importance.

Another relevant contribution lies in the handling of large-scale meteorological datasets. Climate research frequently encounters challenges related to data volume, computational cost, and data quality. While studies such as Ardabili *et al.* [2] highlight the need for efficient computational strategies and robust data processing pipelines, the present study addresses these issues by employing batch processing, reducing numerical precision, and systematically aggregating historical climate records. The resulting preprocessing pipeline consolidates more than two decades of meteorological observations into structured datasets suitable for downstream analysis.

Overall, the findings indicate that LLM agents can function as an interpretive layer over previously processed and aggregated climate data. In the evaluated cases, their contribution consisted of contextualizing candidate temperature anomalies, organizing temporal comparisons, identifying when the available variables were insufficient for a reliable conclusion, and consolidating specialized outputs into a structured report. These functions may improve the accessibility and organization of analytical results, particularly when long temporal series must be communicated to human analysts. However, the agents did not independently calculate the underlying trends, validate extreme climate events, or replace statistical and climatological models. Therefore, the evidence supports

the use of the proposed framework for assisted interpretation and report generation rather than autonomous climate inference.

6 Conclusions

Understanding long-term climate dynamics has become increasingly important as global temperature patterns continue to change and extreme weather events become more frequent. Reliable analysis of historical meteorological data is essential for identifying trends, detecting anomalies, and supporting climate-related decision-making. However, the large volume, heterogeneity, and quality issues commonly present in climate datasets create substantial technical challenges for large-scale analysis.

This study proposed a pipeline for large-scale processing and interpretation of climate data using meteorological records and LLM agents. Historical data from the INMET were collected, preprocessed, and aggregated to reduce noise and computational complexity. The resulting datasets were analyzed through a multi-agent LLM framework designed to interpret temperature trends and identify relevant climate patterns over time.

The processed temperature series exhibited an overall upward tendency across the analyzed period. Based on these aggregated inputs, the LLM agents contributed by screening candidate temperature anomalies, organizing trends across predefined temporal intervals, and identifying situations in which the available evidence was insufficient for a reliable meteorological interpretation. In particular, the heat-wave agent contextualized deviations from the historical September baseline, while the cold-wave agent indicated that additional atmospheric variables were necessary before a cold-front classification could be made. These results show that the proposed framework can assist with the interpretation, synthesis, and communication of climate data. They do not, however, demonstrate that LLM agents can independently validate extreme events or generate reliable climate forecasts.

This work contributes to the growing body of research on climate data analysis by proposing an integrated approach that combines large-scale data preprocessing techniques with LLM-based interpretative agents. The proposed pipeline demonstrates how large meteorological datasets can be efficiently processed, summarized, and interpreted, enabling automated climate analysis while maintaining manageable computational requirements. Additionally, the study provides a regional perspective focused on Brazil, contributing to the understanding of climate patterns in a geographically diverse and environmentally significant region.

Despite these contributions, several dataset-related limitations must be acknowledged. Approximately 30% of the raw observations contained invalid sentinel values, such as -9999.0, requiring their conversion to missing values and subsequent treatment. This high proportion of invalid observations may reduce the representativeness of some stations and periods, particularly when missingness is geographically or temporally concentrated. Forward filling assumes

temporal persistence, whereas linear interpolation assumes gradual change; both procedures may smooth abrupt variations and underestimate short-lived extreme events. The IQR-based outlier removal procedure may also have excluded valid climatic extremes, particularly in skewed or seasonally heterogeneous distributions, thereby affecting the estimated frequency and intensity of candidate temperature anomalies. Because the study did not compare alternative imputation and outlier-handling strategies, the robustness of the identified patterns to these preprocessing choices remains uncertain. The aggregation of hourly observations into daily and monthly summaries further reduces local and short-term variability; therefore, the detected warming patterns should be interpreted as broad temporal trends rather than evidence of specific station-level or regional phenomena. Differences in station coverage and data availability over time may also influence the apparent historical trend. Finally, the values reported for 2025–2050 were obtained through trend-based extrapolation rather than through a validated forecasting model and should consequently be interpreted as exploratory scenarios, not as reliable climate predictions.

The study relied on existing instruction-tuned language models and did not include task-specific training or fine-tuning. As a result, the agents' behavior remained dependent on the general capabilities, context limitations, and potential inconsistencies of the selected models. Although the multi-agent organization supported task decomposition and report consolidation, it did not guarantee the correctness of the generated interpretations. Future research should therefore investigate task-specific adaptation, prompt optimization, structured-output constraints, and additional mechanisms to verify that the generated statements are supported by the underlying climate data.

Future research should expand the agent-based analysis by incorporating additional climatic variables already available in the collected dataset, such as humidity, precipitation, atmospheric pressure, and wind patterns, which are essential for more comprehensive meteorological analyses. The integration of regional and global climate models may also improve the reliability of future projections by combining local resolution with broader climatic context. Additionally, the use of machine learning techniques and advanced simulation approaches may further enhance predictive accuracy and enable the identification of complex climate patterns. By improving data quality, expanding model capabilities, and integrating multiple sources of climatic information, future studies will be better equipped to support climate risk mitigation and informed decision-making.

Acknowledgments

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